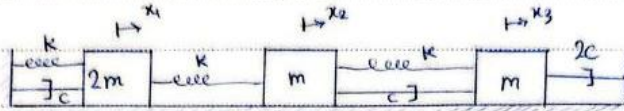




Q1: Lagrange's equation. x_1, x_2, x_3 coordinates



Lagrange's equation: $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) - \left(\frac{\partial L}{\partial x_i} \right) = Q_i$, $L = T - V$, $Q_i = -\frac{\partial R}{\partial x_i}$

$$= \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_i} \right) - \frac{\partial T}{\partial x_i} + \frac{\partial V}{\partial x_i} + \frac{\partial R}{\partial x_i} = 0$$

10 From system can write:

$$T = \frac{1}{2} (2m) \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2 + \frac{1}{2} m \dot{x}_3^2, \quad V = \frac{1}{2} k x_1^2 + \frac{1}{2} k (x_2 - x_1)^2 + \frac{1}{2} k (x_3 - x_2)^2$$

$$R = \frac{1}{2} c \dot{x}_1^2 + \frac{1}{2} c (\dot{x}_3 - \dot{x}_2)^2 + \frac{1}{2} (2c) \dot{x}_3^2$$

For x_1 :

$$x_1: \frac{\partial T}{\partial x_1} = 2m\ddot{x}_1, \quad \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_1} \right) = 2m\ddot{x}_1, \quad \frac{\partial V}{\partial x_1} = kx_1 + k(x_1 - x_2) = 2kx_1 - kx_2, \quad \frac{\partial R}{\partial x_1} = c\dot{x}_1$$

$$15 = 2m\ddot{x}_1 + c\dot{x}_1 + 2kx_1 - kx_2 = 0$$

$$x_2: \frac{\partial T}{\partial x_2} = m\ddot{x}_2, \quad \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_2} \right) = m\ddot{x}_2, \quad \frac{\partial V}{\partial x_2} = k(x_2 - x_1) + k(x_2 - x_3) = 2kx_2 - kx_1 - kx_3, \quad \frac{\partial R}{\partial x_2} = c(\dot{x}_2 - \dot{x}_3)$$

$$= m\ddot{x}_2 + c(\dot{x}_2 - \dot{x}_3) + 2kx_2 - kx_1 - kx_3 = 0$$

$$x_3: \frac{\partial T}{\partial x_3} = m\ddot{x}_3, \quad \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_3} \right) = m\ddot{x}_3, \quad \frac{\partial V}{\partial x_3} = k(x_3 - x_2) = kx_3 - kx_2, \quad \frac{\partial R}{\partial x_3} = c(\dot{x}_3 - \dot{x}_2) + 2c\dot{x}_3$$

$$= m\ddot{x}_3 - c\dot{x}_2 + 3c\dot{x}_3 - kx_2 + kx_3 = 0$$

$$20 \quad 2m\ddot{x}_1 + c\dot{x}_1 + 2kx_1 - kx_2 = 0$$

$$\begin{cases} m\ddot{x}_2 + c(\dot{x}_2 - \dot{x}_3) + 2kx_2 - kx_1 - kx_3 = 0 \\ m\ddot{x}_3 - c\dot{x}_2 + 3c\dot{x}_3 - kx_2 + kx_3 = 0 \end{cases}, \quad M\ddot{x} + C\dot{x} + Kx = 0$$

$$m\ddot{x}_3 - c\dot{x}_2 + 3c\dot{x}_3 - kx_2 + kx_3 = 0$$

$$= \begin{bmatrix} 2m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{bmatrix} + \begin{bmatrix} c & 0 & 0 \\ 0 & c & -c \\ 0 & -c & 3c \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} + \begin{bmatrix} 2k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & k \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Q2: For falling mass (m), Lagrange function is: $L = T - V = m(\frac{1}{2}\dot{x}^2 + gx) + \text{cte}$

and $\int_{t_1}^{t_2} L dt = 0$ proper equation.

$\uparrow \ddot{x}$
 $\downarrow \dot{x}$
 $\downarrow x$

$$T = \frac{1}{2} m \dot{x}^2, \quad V = mgy$$

$y = -x$ $\Rightarrow V = mg(-x) = -mgx$, initial point can be anywhere $\Rightarrow V = -mgx + C$

$$L = T - V = \frac{1}{2} m \dot{x}^2 - (-mgx + C) = \frac{1}{2} m \dot{x}^2 + mgx - C \stackrel{-C=\text{cte}}{=} \frac{1}{2} m \dot{x}^2 + mgx + \text{const}$$

$$\Rightarrow h = m(\frac{1}{2}\dot{x}^2 + gx) + \text{const}$$

For second part from $\int_{t_1}^{t_2} L dt = 0$ we know: $\frac{d}{dt}(\frac{\partial L}{\partial \dot{x}}) - \frac{\partial L}{\partial x} = 0$

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{x}}) = \frac{d}{dt}(m\dot{x}) = m\ddot{x}, \quad \frac{\partial L}{\partial x} = mg \Rightarrow m\ddot{x} - mg = 0 \Rightarrow \ddot{x} = g \quad \text{Newton's law for free fall.}$$

* Constant disappear in derivation and has no effect.

$$Q3: I = \int_1^2 \frac{\sqrt{1+y'^2}}{x} dx \quad y(1)=0, \quad y(2)=1$$

$$L = \sqrt{1+y'^2}/x, \quad \frac{d}{dx}(\frac{\partial L}{\partial y'}) = 0 \Rightarrow \frac{\partial L}{\partial y'} = C = \frac{1}{x} \frac{y'}{\sqrt{1+y'^2}} \Rightarrow \frac{y'}{\sqrt{1+y'^2}} = Cx$$

$$\frac{y'^2}{1+y'^2} = C^2 x^2 \Rightarrow y'^2 = \frac{C^2 x^2}{1-C^2 x^2} \Rightarrow y' = \frac{\pm Cx}{\sqrt{1-C^2 x^2}}$$

$$y' = \frac{dy}{dx} \Rightarrow y = \int \frac{Cx}{\sqrt{1-C^2 x^2}} dx, \quad u = 1-C^2 x^2 \Rightarrow du = -2Cx dx \Rightarrow y = -\frac{1}{C} \sqrt{1-C^2 x^2} + A$$

$$\Rightarrow y = A \mp \frac{1}{C} \sqrt{1-C^2 x^2} \xrightarrow{R=1/C} y = A \mp \sqrt{R^2 - x^2} \xrightarrow{\text{square}} x^2 + (y-A)^2 = R^2 \quad \text{Circle eqn.}$$

$$\begin{cases} y(1)=0 \Rightarrow 1 + (0-A)^2 = R^2 \Rightarrow 1 + A^2 = R^2 \\ y(2)=1 \Rightarrow 4 + (1-A)^2 = R^2 \Rightarrow 5 - 2A + A^2 = R^2 \end{cases} \Rightarrow 1 + A^2 = A^2 - 2A + 5 \Rightarrow A = 2 \Rightarrow R = \sqrt{5}$$

$$\Rightarrow (y-2) = \pm \sqrt{5-x^2}, \quad y(1)=0 \Rightarrow y-2 = 0-2 = -2 \Rightarrow \text{negative}$$

$$\Rightarrow y = 2 - \sqrt{5-x^2}, \quad 1 \leq x \leq 2$$

$$\text{Q4: } I = \int_0^1 (y'^2 - 2xyy' - 2\beta y') dx$$

$$L = y'^2 - 2xyy' - 2\beta y' \quad \frac{\partial L}{\partial y} = -2xy' \quad \frac{\partial F}{\partial y'} = 2y' - 2xy - 2\beta = \frac{d}{dx} \left(\frac{\partial L}{\partial y'} \right) - \frac{\partial L}{\partial y} = 0$$

$$\Rightarrow \frac{d}{dx} (2y' - 2xy - 2\beta) - (-2xy') = 0 \Rightarrow 2y'' - 2xy' + 2xy' = 2y'' = 0 \Rightarrow y'' = 0$$

$$y(x) = C_1 x + C_0$$

$$\text{E: End part: } \frac{\partial F}{\partial y'} = 0, \quad \frac{\partial F}{\partial y'} = 2y' - 2xy - 2\beta$$

$$\text{a) } y(0) = 0, y(1) = 1 \Rightarrow y(0) = C_1(0) + C_0 \Rightarrow C_0 = 0 \Rightarrow y(1) = 1 = C_1 \Rightarrow y(x) = x$$

$$\text{10. b) } y(0) = 0 \Rightarrow C_0 = 0 \Rightarrow y = C_1 x$$

$$\frac{\partial F}{\partial y'} = 2y' - 2xy - 2\beta = 2C_1 - 2dC_1x - 2\beta = 0 \quad x=1 \Rightarrow 2C_1 - 2dC_1 = 2\beta \Rightarrow C_1 = \beta / (1-d)$$

$$y(x) = \frac{\beta}{1-d} x, \quad d \neq 1 \quad (\text{if } d=1 \Rightarrow \beta=0)$$

$$\text{c) } y(1) = 1 \Rightarrow C_1 + C_0 = 1, \quad y = C_1 x + C_0 \quad C_1 = 1 - C_0 \quad \left. \begin{array}{l} 1 - C_0 = dC_0 + \beta \\ C_0 = \frac{1-\beta}{d+1}, C_1 = \frac{d+\beta}{d+1} \end{array} \right\}$$

$$\frac{\partial F}{\partial y'} \Big|_{x=0} = 2y' - 2xy - 2\beta = 2C_1 - 2dC_0 - 2\beta = 0 \Rightarrow C_1 = dC_0 + \beta$$

$$\text{15. } y(x) = \frac{d+\beta}{1+d} x + \frac{1-\beta}{1+d}, \quad d \neq -1 \quad (\text{if } d=-1 \Rightarrow \beta=1)$$

$$\text{d) } \frac{\partial F}{\partial y'} = 0 \Rightarrow 2y' - 2xy - 2\beta = 0, \quad y = C_1 x + C_0$$

$$\left. \begin{array}{l} x=0 \Rightarrow 2C_1 - 2dC_0 - 2\beta = 0 \Rightarrow C_1 = dC_0 + \beta \\ x=1 \Rightarrow 2C_1 - 2d(C_1 + C_0) - 2\beta = 0 \Rightarrow C_1 = d(C_1 + C_0) + \beta \end{array} \right\} \Rightarrow dC_0 + \beta = d(C_1 + C_0) + \beta$$

$$dC_1 = 0 \Rightarrow d=0 \text{ or } C_1=0$$

$$\text{20. } C_1 = 0 \Rightarrow C_1 = -\beta/d \Rightarrow y = -\beta/d x$$

$$d=0 \Rightarrow C_1 = \beta \Rightarrow y = \beta x + C_0$$

Q5: wire, move from O to P_2 in least time.

$$mgy + \frac{1}{2}mv^2 = cte = 0 \Rightarrow mgy = -\frac{1}{2}mv^2 \Rightarrow mgy = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gy}$$

$$ds = \sqrt{1+y'^2} dx, \quad dt = \frac{ds}{v} = \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} dx \Rightarrow T = \int_0^{x_2} \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} dx \quad y(0)=0, y(x_2)=y_2$$

$$\Rightarrow L = \sqrt{1+y'^2} / \sqrt{2gy}, \quad L - y' \frac{\partial L}{\partial y'} = C, \quad \frac{\partial L}{\partial y'} = \frac{1}{\sqrt{2gy}} \cdot \frac{y'}{\sqrt{1+y'^2}}$$

$$F - y' \frac{\partial L}{\partial y'} = \frac{1}{\sqrt{2gy}} \left(\sqrt{1+y'^2} - \frac{y'^2}{\sqrt{1+y'^2}} \right) = \frac{1}{\sqrt{2gy}} \frac{1}{\sqrt{1+y'^2}} = C \Rightarrow \sqrt{1+y'^2} \sqrt{2gy} = 1/C$$

Square

$$\Rightarrow (1+y'^2)y = cte = 2a \Rightarrow (1+y'^2)y = 2a \Rightarrow 1+y'^2 = \frac{2a}{y} \Rightarrow \frac{dy}{dx} = \sqrt{\frac{2a-y}{y}}$$

$$\Rightarrow dx = \sqrt{\frac{y}{2a-y}} dy, \quad y = a(1-\cos\theta), \quad 2a-y = 2a-a+\cos\theta = a(1+\cos\theta)$$

$$\Rightarrow \frac{y}{2a-y} = \frac{1-\cos\theta}{1+\cos\theta} = \tan^2(\theta/2) \Rightarrow \sqrt{\frac{y}{2a-y}} = \tan(\theta/2), \quad dy = a \sin\theta d\theta$$

$$\Rightarrow dx = \tan(\theta/2) a \sin\theta d\theta, \quad \sin\theta = 2\sin(\theta/2)\cos(\theta/2), \quad \tan(\theta/2) = \frac{\sin(\theta/2)}{\cos(\theta/2)}$$

$$\Rightarrow \tan(\theta/2) \sin\theta = 2\sin^2(\theta/2) = 1-\cos\theta \Rightarrow dx = a(1-\cos\theta) d\theta$$

$$x = \int a(1-\cos\theta) d\theta = a(\theta - \sin\theta) + C, \quad \theta(x=0)=0 \Rightarrow C=0 \Rightarrow x = a(\theta - \sin\theta)$$

$$y = a(1-\cos\theta)$$

$$\Rightarrow \begin{cases} x(\theta) = a(\theta - \sin\theta) \\ y(\theta) = a(1-\cos\theta) \end{cases}$$

$$\cos\theta = 1 - y/a$$

$$y/a = 1 - \cos\theta \Rightarrow \theta = \arccos(1 - y/a)$$

$$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - (1 - y/a)^2} = \sqrt{\frac{2y}{a} - \left(\frac{y}{a}\right)^2}$$

$$\Rightarrow x = a(\theta - \sin\theta) = a \left(\arccos(1 - y/a) - \sqrt{\frac{2y}{a} - \left(\frac{y}{a}\right)^2} \right)$$

$$Q6: J(n) = \int_0^{n/2} [\dot{x}_1^2(t) + \dot{x}_2^2(t) + 2x_1(t)x_2(t)] dt$$

$$L = \dot{x}_1^2 + \dot{x}_2^2 + 2x_1x_2 \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) - \frac{\partial L}{\partial x_i} = 0$$

$$x_1: \frac{d}{dt}(2\dot{x}_1) - 2x_2 = 0 \Rightarrow \ddot{x}_1 = x_2, \quad x_2: \frac{d}{dt}(2\dot{x}_2) - 2x_1 = 0 \Rightarrow \ddot{x}_2 = x_1$$

$$\Rightarrow x_1 - x_2 = 0 \Rightarrow x_1 = A \cosh t + B \sinh t + C \cosh t + D \sinh t$$

$$x_2 = A \cosh t + B \sinh t - C \cosh t - D \sinh t$$

$$\begin{cases} x_1(0) = 0 \Rightarrow A + C = 0 \Rightarrow A = -C \\ x_2(0) = 0 \Rightarrow A - C = 0 \Rightarrow A = C \end{cases} \Rightarrow A = C = 0 \Rightarrow \begin{cases} x_1^{(1)} = B \sinh t + D \cosh t \\ x_2^{(1)} = B \sinh t - D \cosh t \end{cases}$$

$$a) \theta = [5t+3, 1/2 t^2]^T \Rightarrow x(t) = \theta(t)$$

$$\Rightarrow \begin{cases} B \sinh t + D \cosh t = 5t+3 \\ B \sinh t - D \cosh t = t^2/2 \end{cases} \Rightarrow \begin{cases} B = (5t+3 + t^2/2) / 2 \sinh t \\ D = (5t+3 - t^2/2) / 2 \sinh t \end{cases}$$

$$\Rightarrow L + \frac{\partial L}{\partial x} (\dot{\theta}(t) \cdot \dot{x}) = 0, \quad \dot{\theta}(t) = [5, t]^T, \quad \frac{\partial L}{\partial x} = 2\dot{x}_1 + 2\dot{x}_2$$

$$\Rightarrow -\dot{x}_1(t)^2 - \dot{x}_2(t)^2 + 2x_1(t)\dot{x}_2(t) + 10\dot{x}_1(t) + 2t\dot{x}_2(t) = 0$$

$$x_1(t) = B \cosh t + D \cosh t, \quad x_2(t) = B \cosh t - D \cosh t, \quad x_1(t) = 5t+3, \quad x_2(t) = t^2/2$$

$$\Rightarrow -(B \cosh t + D \cosh t)^2 - (B \cosh t - D \cosh t)^2 + t^2(5t+3) + 10(B \cosh t + D \cosh t) + 2t(B \cosh t - D \cosh t) = 0$$

$$\Rightarrow B = (5t+3) + t^2/2, \quad D = (5t+3) - t^2/2$$

$$\text{calculator} \quad t = 0.582 \Rightarrow \begin{cases} x_1(t) = 4.938 \sinh t + 5.220 \cosh t \\ x_2(t) = 4.938 \sinh t - 5.220 \cosh t \end{cases} \quad t = 0.582$$

$$b) x_1(t) + 3x_2(t) + 5t = 15 \Rightarrow \varphi(x(t)) = x_1(t) + 3x_2(t) + 5t = 15 \Rightarrow t = t, \quad \varphi = x_1, 3x_2, 5t$$

$$\frac{\partial L}{\partial x} = \mu \varphi_x, \quad L = \dot{x}_1^2 + \dot{x}_2^2 + 2x_1x_2 \Rightarrow \mu \varphi_t = 0 \Rightarrow 2\dot{x}_1(t) = \mu, \quad 2\dot{x}_2(t) = 3\mu \Rightarrow \dot{x}_2 = 3\dot{x}_1$$

$$\Rightarrow \ddot{x}_2 = 3\ddot{x}_1 \Rightarrow B \cosh t + D \cosh t = 3(B \cosh t + D \cosh t) \Rightarrow D = B \cosh t / 2 \cosh t$$

$$L = \dot{x}_1^2 + \dot{x}_2^2 + 2x_1x_2 = (\dot{x}_1^2 + 9\dot{x}_1^2) + 2x_1x_2 = 10\dot{x}_1^2 + 2x_1x_2 = 5\mu = 10\dot{x}_1$$

$$x_1(t) + 3x_2(t) + 5t = 15 \Rightarrow (B \sinh t + D \sinh t) + 3(B \sinh t - D \sinh t) + 5t = 15$$

$$\Rightarrow 4B \sinh t - 2D \sinh t + 5t = 15$$

$$\text{By compute } \star, \star\star, \star\star\star \text{ and use solver: } t = 4.392, \quad B = -0.024, \quad D = -1.575$$

$$\begin{cases} x_1(t) = -0.024 \sinh t - 1.575 \cosh t \\ x_2(t) = -0.024 \sinh t + 1.575 \cosh t \end{cases} \quad t \leq 4.392$$

$$Q7a: \mathcal{J}(\omega) = \int_{t_0}^{t_f} [\dot{\omega}_1(t) + \dot{\omega}_2(t) \omega_2(t) + \dot{\omega}_2(t) + \dot{\omega}_3(t)] dt$$

$$\dot{\omega}_1(t) = \omega_2(t), \quad \dot{\omega}_2(t) = -\omega_2(t) + [1 - \omega_1^2(t)] \omega_2(t) + \omega_3(t)$$

$$\tilde{L} = \omega_1^2 + \omega_1 \omega_2 + \omega_2^2 + \omega_3^2$$

$$L = \tilde{L} + \lambda_1 g_1 + \lambda_2 g_2 = \omega_1^2 + \omega_1 \omega_2 + \omega_2^2 + \omega_3^2 + \lambda_1 (\dot{\omega}_1 - \omega_2) + \lambda_2 (\dot{\omega}_2 - \omega_1 - (1 - \omega_1^2) \omega_2 - \omega_3)$$

$$\frac{\partial L}{\partial \omega_1} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\omega}_1} \right) = 0$$

$$\omega_1: \frac{\partial L}{\partial \omega_1} = 2\omega_1 + \omega_2 + \lambda_2(-1 + 2\omega_1 \omega_2), \quad \frac{\partial L}{\partial \dot{\omega}_1} = \lambda_1 \Rightarrow \frac{d}{dt} \frac{\partial L}{\partial \dot{\omega}_1} = \dot{\lambda}_1$$

$$\Rightarrow \dot{\lambda}_1 = 2\omega_1 + \omega_2 + \lambda_2(-1 + 2\omega_1 \omega_2)$$

$$\omega_2: \frac{\partial L}{\partial \omega_2} = \omega_1 + 2\omega_2 - \lambda_1 - \lambda_2(1 - \omega_1^2), \quad \frac{\partial L}{\partial \dot{\omega}_2} = \lambda_2 \Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\omega}_2} \right) = \dot{\lambda}_2$$

$$\Rightarrow \dot{\lambda}_2 = \omega_1 + 2\omega_2 - \lambda_1 - \lambda_2(1 - \omega_1^2)$$

$$\omega_3: \frac{\partial L}{\partial \omega_3} = 2\omega_3 - \lambda_2, \quad \frac{\partial L}{\partial \dot{\omega}_3} = 0 \Rightarrow 2\omega_3 - \lambda_2 = 0 \Rightarrow \lambda_2 = 2\omega_3$$

$$\Rightarrow \begin{cases} \dot{\omega}_1 = \omega_2 \\ \dot{\omega}_2 = -\omega_1 + (1 - \omega_1^2) \omega_2 + \omega_3 \end{cases} \quad \begin{cases} \dot{\lambda}_1 = 2\omega_1 + \omega_2 + \lambda_2(-1 + 2\omega_1 \omega_2) \\ \dot{\lambda}_2 = \omega_1 + 2\omega_2 - \lambda_1 - \lambda_2(1 - \omega_1^2) \end{cases} \quad \lambda_2 = 2\omega_3$$

$$Q8: x(t) = t^2 + 2, \quad x(-2) = 0, \quad x(1) = 0$$

$$ds^2 = dt^2 + dx^2 \Rightarrow ds = \sqrt{1 + \dot{x}^2} dt \Rightarrow \text{Length} = \int_{-2}^1 \sqrt{1 + \dot{x}^2} dt \Rightarrow \mathcal{J}(x) = \int_{-2}^1 \sqrt{1 + \dot{x}^2} dt$$

$$L = \sqrt{1 + \dot{x}^2}, \quad \frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = 0$$

$$\Rightarrow \frac{\partial L}{\partial x} = \frac{\dot{x}}{\sqrt{1 + \dot{x}^2}} \Rightarrow \frac{d}{dt} \left(\frac{\dot{x}}{\sqrt{1 + \dot{x}^2}} \right) = 0 \Rightarrow \frac{\dot{x}}{\sqrt{1 + \dot{x}^2}} = cte \Rightarrow \dot{x} \text{ is const.} \Rightarrow x(t) \text{ straight line}$$

$$P = (-5, 5+2), \quad S \in [-2, 1] \Rightarrow \text{Circ: } AP + PB \Rightarrow L(S) = |AP| + |PB|$$

$$\Rightarrow L(S) = \sqrt{(S+2)^2 + (S+2)^2} + \sqrt{(S-1)^2 + (S+2)^2} \quad \min L(S)$$

$$\Rightarrow L'(S) = (2S^3 + 5S + 2)/I + (2S^3 + 5S - 1)/II = 0 \xrightarrow{\text{Calculator}} S = -0.0695 \Rightarrow P = (-0.0695, 2.005)$$

$$\begin{cases} AP: m_1 = (S+2-0)/(S-(-2)) = (S+2)/(S+2) \Rightarrow x(t) = m_1(t+2) \Rightarrow x(t; S, m_1 = 1.0385 \end{cases}$$

$$\begin{cases} PB: m_2 = 0 - (S+2)/(1-S) = -(S+2)/(1-S) \Rightarrow x(t) = m_2(t-1) \Rightarrow x(t; 1, m_2 = -1.8745 \end{cases}$$

$$x(t) = \begin{cases} 1.0385(t+2) & -2 \leq t \leq -0.0695 \\ -1.8745(t-1) & -0.0695 \leq t \leq 1 \end{cases} \quad L_{\min} = L(S = -0.0695) = 5.0554 \text{ (unit)}$$